

Postal Tracking System Database Design and Implementation

Introduction

Postal Tracking Systems are widely used to monitor the movement and delivery status of packages throughout the shipping process. A well-designed database is essential for storing customer information, package details, tracking records, and postal office operations efficiently.

This project focuses on the design and implementation of a relational database for a **Postal Tracking System** using **MySQL** and **MySQL Workbench**. The system enables users to register packages, track shipment progress using unique tracking codes, and maintain a complete history of package status updates.

System Requirements

Functional Requirements

- Register customer information
- Register sender information
- Register receiver information
- Register package information
- Generate and manage unique tracking codes
- Record package status updates
- Manage postal office information
- Store package tracking history
- Search packages by tracking code
- Display the latest package status

Non-Functional Requirements

- Prevent duplicate records
- Maintain data integrity and consistency
- Provide efficient data retrieval
- Support future scalability
- Ensure data security

Database Design

The database is composed of the following entities:

Entity
Customer
Package
Sender
Receiver
Status
PostOffice
TrackingHistory

Entity Descriptions

Customer

Stores customer information.

```
Customer(  
    CustomerID,  
    FullName,
```

```
    Phone,  
    Address  
)
```

Package

Stores package information and tracking details.

```
Package(  
    PackageID,  
    TrackingCode,  
    Weight,  
    SendDate,  
    CustomerID  
)
```

Sender

Stores sender information associated with a package.

```
Sender(  
    SenderID,  
    Name,  
    Phone,  
    Address,  
    PackageID  
)
```

Receiver

Stores receiver information associated with a package.

```
Receiver(  
    ReceiverID,  
    Name,  
    Phone,
```

```
    Address,  
    PackageID  
)
```

Status

Stores available package statuses.

```
Status(  
    StatusID,  
    StatusName  
)
```

PostOffice

Stores postal office information.

```
PostOffice(  
    OfficeID,  
    OfficeName,  
    City  
)
```

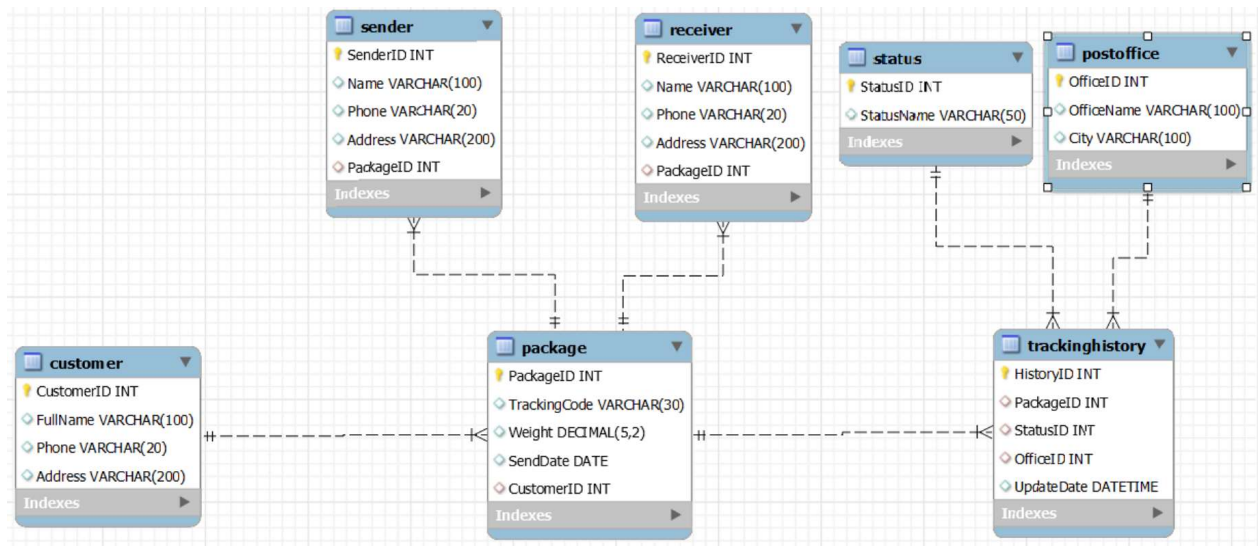
TrackingHistory

Stores package tracking history and status updates.

```
TrackingHistory(  
    HistoryID,  
    PackageID,  
    StatusID,  
    OfficeID,  
    UpdateDate  
)
```

ERD

The Entity Relationship Diagram (ERD) was designed using **MySQL Workbench**. The diagram illustrates relationships between customers, packages, senders, receivers, statuses, postal offices, and tracking history records.



Normalization

First Normal Form (1NF)

All attributes contain atomic values, and no repeating groups or multi-valued attributes exist within any table. Therefore, all tables satisfy the requirements of **1NF**.

Second Normal Form (2NF)

Each table uses a primary key, and all non-key attributes are fully dependent on the entire primary key. No partial dependencies exist. Therefore, all tables satisfy **2NF**.

Third Normal Form (3NF)

There are no transitive dependencies among non-key attributes. Every non-key attribute depends directly on the primary key. Therefore, all tables satisfy **3NF**.

Boyce-Codd Normal Form (BCNF)

A review of the database schema shows that every determinant is a candidate key. No dependency violating BCNF exists; therefore, the database also satisfies **BCNF**.

Software Environment

Component	Technology
DBMS	MySQL
Database Design Tool	MySQL Workbench
Query Language	SQL
Operating System	Windows 11

Conclusion

A relational database for a Postal Tracking System was successfully designed and implemented. The database structure supports customer management, package registration, sender and receiver information, postal office operations, package status tracking, and historical tracking records.

By applying normalization up to **BCNF**, data redundancy was minimized while maintaining consistency and integrity. The resulting schema is scalable, efficient, and provides a solid foundation for future development of a complete postal tracking application.